Vectors I Basics Worksheet: Questions

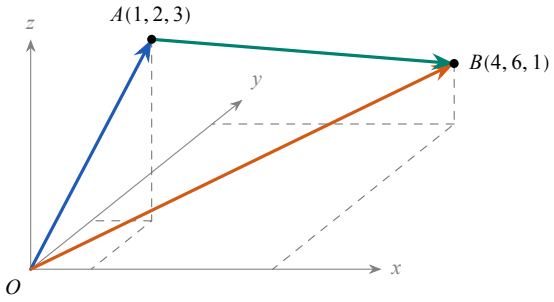
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Basics of vectors: **i**, **j**, **k** notation, magnitude, unit vectors, parallel vectors, collinear points, the ratio theorem, the scalar product and the vector product.

1 **i**, **j**, **k** Notation and Column Vectors

1 Write $\mathbf{a}=3\mathbf{i}-2\mathbf{j}+5\mathbf{k}$ as a column vector, and write $\mathbf{b}=\begin{pmatrix}4\\0\\-1\end{pmatrix}$ in terms of **i**, **j** and **k**.

2 The points A and B have coordinates $(1,2,3)$ and $(4,6,1)$. Find $\overrightarrow{AB}$ in terms of **i**, **j** and **k**.



3 Given $\mathbf{p}=2\mathbf{i}+\mathbf{k}$ and $\mathbf{q}=-\mathbf{i}+3\mathbf{j}$, find **r** such that $2\mathbf{p}-3\mathbf{q}+\mathbf{r}=\mathbf{0}$. Give **r** in terms of **i**, **j** and **k**.

4 For the following statements, state whether it is **true** or **false**.

(a) $\mathbf{i}+\mathbf{j}+\mathbf{k}=\begin{pmatrix}1\\1\\1\end{pmatrix}$

(b) The **j**-component of $4\mathbf{i}-\mathbf{j}+2\mathbf{k}$ is -1

(c) The position vector of $P(1,-2,0)$ is $\mathbf{i}-2\mathbf{j}$

(d) $3\mathbf{i}+2\mathbf{k}=\begin{pmatrix}3\\2\\0\end{pmatrix}$

(e) $\overrightarrow{AB}=\overrightarrow{OA}-\overrightarrow{OB}$

2 Magnitude

1 Find $|2\mathbf{i}-3\mathbf{j}+6\mathbf{k}|$.

2 The points A and B have coordinates $(1, 2, -1)$ and $(3, -4, 2)$. Find the distance AB.

3 Given $\mathbf{a}=\begin{pmatrix} t \\ 2 \\ t-1 \end{pmatrix}$ and $|\mathbf{a}|=3$, find the possible values of t.

4 For the following statements, state whether it is **true** or **false**.

(a) $|-\mathbf{a}|=|\mathbf{a}|$ for every vector $\mathbf{a}$

(b) $|3\mathbf{a}|=3|\mathbf{a}|$

(c) $\left|\begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix}\right|=-5$

(d) $\left|\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right|=1$

(e) $|\mathbf{i}+2\mathbf{j}-2\mathbf{k}|=1+2-2=1$

(f) $|\mathbf{a}+\mathbf{b}|=|\mathbf{a}|+|\mathbf{b}|$ for all vectors $\mathbf{a}$, $\mathbf{b}$

(g) $|\mathbf{a}+\mathbf{b}|\leq|\mathbf{a}|+|\mathbf{b}|$ for all vectors $\mathbf{a}$, $\mathbf{b}$

(h) For all $\lambda\in\mathbb{R}$, $|\lambda\mathbf{a}|=\lambda|\mathbf{a}|$

3 Unit Vectors

1 Find the unit vector in the direction of $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

2 Find the vector of magnitude 10 in the direction of $\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$.

3 The points A and B have coordinates $(1, 0, 2)$ and $(3, -6, 5)$. Find the vector of magnitude 14 in the direction *opposite* to $\overrightarrow{AB}$.

4 For the following statements, state whether it is **true** or **false**.

(a) $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is a unit vector

(b) $\frac{1}{3}\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ is a unit vector

(c) $\mathbf{i}+\mathbf{j}$ is a unit vector

(d) $|\hat{\mathbf{a}}| = 1$ for every non-zero vector $\mathbf{a}$

(e) $\mathbf{a}$ and $\hat{\mathbf{a}}$ are parallel

4 Parallel Vectors ($\mathbf{a} = \lambda \mathbf{b}$)

1 Show that $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -6 \\ 3 \\ -9 \end{pmatrix}$ are parallel.

2 Find p and q such that $\begin{pmatrix} 4 \\ p \\ -2 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 3 \\ q \end{pmatrix}$.

3 Find all vectors of magnitude 6 that are parallel to $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$.

4 For the following statements, state whether it is **true** or **false**.

(a) If $\mathbf{a} \parallel \mathbf{b}$, then $\mathbf{a} = \mathbf{b}$

(b) $\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix}$

(c) $\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$

(d) $\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix}$ is parallel to $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$

(e) If $\mathbf{a} = -2\mathbf{b}$, then $\mathbf{a}$ and $\mathbf{b}$ are not parallel since they point in opposite directions

5 Collinear Points

1 Show that the points $A(1, 2, 3)$, $B(3, 5, 4)$ and $C(7, 11, 6)$ are collinear.

2 For the points in Question 1, find the ratio $AB : BC$.

3 The points $P(2, -1, 0)$, $Q(4, 1, 3)$ and $R(a, 7, b)$ are collinear. Find a and b.

4 For the following statements, state whether it is **true** or **false**.

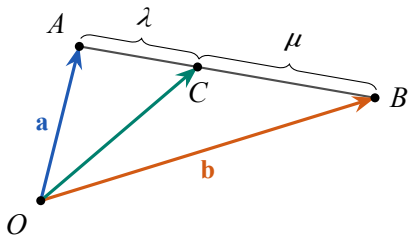
(a) If $\overrightarrow{AB} \parallel \overrightarrow{CD}$, then A, B, C and D are collinear

(b) If $\overrightarrow{AB} = 3\overrightarrow{AC}$, then A, B and C are collinear

(c) $A(0, 0, 0)$, $B(1, 1, 1)$ and $C(2, 2, 3)$ are collinear

(d) The points with position vectors $\mathbf{a}$, $2\mathbf{a}$ and $-\mathbf{a}$ are collinear ($\mathbf{a} \neq \mathbf{0}$)

6 Ratio Theorem



In the True/False statements, C lies on AB with $AC:CB=\lambda:\mu$, $\overrightarrow{OA}=\mathbf{a}$ and $\overrightarrow{OB}=\mathbf{b}$.

1 Let $\overrightarrow{OA}=\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\overrightarrow{OB}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$. Given that C lies on AB, find the coordinates of C if

(a) $AC:CB=1:2$,

(b) $3AC=2CB$,

(c) $2AB=5CB$,

(d) Given instead that C lies on AB produced, and $3AC=7BC$.

2 Find the position vector of the midpoint M of AB, where A and B have coordinates $(3,-1,5)$ and $(-1,5,1)$.

3 For the following statements, state whether it is **true** or **false**.

(a) $\overrightarrow{OC}=\dfrac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$

(b) $\overrightarrow{OC}=\dfrac{\lambda\mathbf{a}+\mu\mathbf{b}}{\lambda+\mu}$

(c) If $\lambda:\mu=1:1$, then $\overrightarrow{OC}=\dfrac{1}{2}(\mathbf{a}+\mathbf{b})$

(d) If $\lambda:\mu=1:3$, then $\overrightarrow{OC}=\dfrac{3\mathbf{a}+\mathbf{b}}{4}$

(e) If $\lambda:\mu=2:1$, then $\overrightarrow{OC}=\dfrac{2\mathbf{a}+\mathbf{b}}{3}$

7 Scalar Product

1 Properties of scalar product

It is given that $\mathbf{a}=\begin{pmatrix}1\\2\\3\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}-3\\2\\7\end{pmatrix}$, $\mathbf{c}=\begin{pmatrix}4\\-5\\-1\end{pmatrix}$. Find the following

(a) $\mathbf{a}\cdot\mathbf{b}$ (b) $\mathbf{b}\cdot\mathbf{a}$ (c) $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})$ (d) $\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$

(e) $\mathbf{a}\cdot(2\mathbf{c})$ (f) $(2\mathbf{a})\cdot\mathbf{c}$ (g) $\mathbf{a}\cdot\mathbf{a}$

2 Scalar product formula: $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\theta$

(a) Points O, A, B are such that $\overrightarrow{OA}=\mathbf{i}+4\mathbf{j}-3\mathbf{k}$ and $\overrightarrow{OB}=5\mathbf{i}-\mathbf{j}$

(i) Find the size of angle AOB

(ii) Find the size of angle OAB.

(b) It is given that $\mathbf{c}=\begin{pmatrix}2\\3\\6\end{pmatrix}$, $\mathbf{d}=\begin{pmatrix}1\\-2\\3\end{pmatrix}$ and $\mathbf{e}=\begin{pmatrix}3\\2\\-2\end{pmatrix}$. Find the angle between the vectors $\mathbf{c}$ and

(i) the vector $\mathbf{d}$

(ii) the vector $\mathbf{e}$

(iii) the positive y-axis

(c) It is given that two vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy the following relationship:

$$|\mathbf{a}||\mathbf{b}|=6.$$

Find the value of $\mathbf{a}\cdot\mathbf{b}$ for each of the following scenarios.

(i) The angle between $\mathbf{a}$ and $\mathbf{b}$ is $60°$.

(ii) The angle between $\mathbf{a}$ and $\mathbf{b}$ is $120°$.

(iii) Vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

3 Perpendicularity of vectors

(a) Referred to the origin O, the position vectors of the points A and B are $\mathbf{i}-\mathbf{j}+2\mathbf{k}$ and $2\mathbf{i}+4\mathbf{j}+\mathbf{k}$ respectively. Show that OA is perpendicular to OB.

(b) The position vectors $\mathbf{a}$ and $\mathbf{b}$ are given by

$$\mathbf{a}=2p\mathbf{i}+3p\mathbf{j}+6p\mathbf{k} \quad \text{and} \quad \mathbf{b}=\mathbf{i}-2\mathbf{j}+2\mathbf{k}$$

where $p>0$. It is given that $|\mathbf{a}|=|\mathbf{b}|$.

(i) Find the exact value of p.

(ii) Show that the vectors $(\mathbf{a}+\mathbf{b})$ and $(\mathbf{a}-\mathbf{b})$ are perpendicular.

4 Length of projection

(a) Given that $\mathbf{a}=2\mathbf{i}+\mathbf{j}+2\mathbf{k}$ and $\mathbf{b}=-\mathbf{i}+2\mathbf{k}-\mathbf{k}$, find the length of projection and the projection vector of

(i) $\mathbf{a}$ onto $\mathbf{b}$,

(ii) $\mathbf{b}$ onto $\mathbf{a}$.

(b) With reference to the origin O, the points A, B and C have position vectors given by

$$\mathbf{a}=\mathbf{i}+2\mathbf{j}-\mathbf{k},\ \mathbf{b}=3\mathbf{i}+\mathbf{k} \text{ and } \mathbf{c}=2\mathbf{i}+2\mathbf{j}+3\mathbf{k}$$

respectively. Find the length of projection of $\overrightarrow{BA}$ onto $\overrightarrow{BC}$.

5 For the following statements, state whether it is **true** or **false**.

(a) $\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$.

(b) $\mathbf{a}\cdot\mathbf{a}=|\mathbf{a}|^2$.

(c) $\mathbf{i}\cdot\mathbf{j}=0$.

(d) $\mathbf{a}\cdot\mathbf{b}$ is a vector.

(e) $\mathbf{i}\cdot\mathbf{j}=\mathbf{k}$.

(f) If $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$, then $\mathbf{a}\cdot\mathbf{b}=0$.

(g) $(2\mathbf{a})\cdot(3\mathbf{b})=6\,\mathbf{a}\cdot\mathbf{b}$.

(h) $(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ is a vector.

(i) $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\theta$, where θ is the non-reflex angle between $\mathbf{a}$ and $\mathbf{b}$.

(j) $\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=\begin{pmatrix}2\\6\\15\end{pmatrix}$.

(k) If $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$.

(l) $|\mathbf{a}\cdot\mathbf{b}|=|\mathbf{a}||\mathbf{b}|$.

(m) $\mathbf{a}\cdot\mathbf{b}\le|\mathbf{a}||\mathbf{b}|$.

(n) For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}>0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is acute.

(o) For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}<0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is obtuse.

(p) For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}\ge0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is acute.

(q) $\mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is a scalar.

(r) $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=|\mathbf{a}|^2+|\mathbf{b}|^2$.

(s) For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|$, then $\mathbf{a}$ and $\mathbf{b}$ are parallel.

(t) $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\theta$ also holds if θ is the reflex angle between $\mathbf{a}$ and $\mathbf{b}$.

(u) Let θ be the angle ABC. Then $\overrightarrow{BA}\cdot\overrightarrow{BC}=|\overrightarrow{BA}||\overrightarrow{BC}|\cos\theta$.

(v) The length of projection of $\mathbf{a}$ onto $\mathbf{b}$ is equal to the length of projection of $\mathbf{b}$ onto $\mathbf{a}$.

(w) If $|\mathbf{a}|=|\mathbf{b}|$, then $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=0$.

(x) $\mathbf{a}\cdot\mathbf{b}=\mathbf{a}\cdot\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$.

(y) Let θ be the angle ABC. Then $\overrightarrow{AB}\cdot\overrightarrow{BC}=|\overrightarrow{AB}||\overrightarrow{BC}|\cos\theta$.

(z) Let θ be the angle CBD. Then $\overrightarrow{BD}\cdot\overrightarrow{CB}=|\overrightarrow{BD}||\overrightarrow{CB}|\cos\theta$.

8 Vector Product

1 Properties of vector product

(a) Find $\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \times \begin{pmatrix} 5 \\ 7 \\ 1 \end{pmatrix}$

(b) Given that $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$, find

(i) $\mathbf{a} \times \mathbf{b}$; (ii) $\mathbf{b} \times \mathbf{a}$; (iii) $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$; (iv) $\mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$;

(v) $\mathbf{a} \times (2\mathbf{b})$; (vi) $(2\mathbf{a}) \times \mathbf{b}$; (vii) $\mathbf{a} \times \mathbf{a}$;

2 Parallel vectors

(a) Three non-zero vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy $\mathbf{a} \times 3\mathbf{b} = 2\mathbf{a} \times \mathbf{c}$. Show that $\mathbf{a}$ and $3\mathbf{b} - 2\mathbf{c}$ are parallel.

(b) Find the scalar product and vector product of the following vectors, stating whether they are parallel, perpendicular, or neither.

(i) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$

(ii) $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + \mathbf{k}$

(iii) $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ -8 \\ 4 \end{pmatrix}$

3 Vector product formula

Using the Vector Product, find the acute angles between the following vectors:

(a) $\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$ (b) $\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}$ (c) $\begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$

4 Perpendicular distance

(a) Relative to an origin O, two point A and B have position vectors given by $\mathbf{a} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = -2\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$. Find the following:

(i) The shortest distance from A to OB

(ii) The shortest distance from B to OA

(b) Relative to the origin O, two points A and B have position vectors given by $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ respectively.

(i) Find the length of projection of $\mathbf{b}$ onto $\mathbf{a}$.

(ii) Hence, or otherwise, find the shortest distance from B to OA.

(c) Relative to the origin O, the points A, B and C have position vectors given by $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = \mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{c} = 3\mathbf{i} - \mathbf{j} + \mathbf{k}$. Find the shortest distance from C to the line joining A and B.

(d) With respect to an origin O, the points A and B have position vectors $\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ and $5\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ respectively. F is the foot of perpendicular of point B on the line that passes through A and is parallel to the vector $\mathbf{d} = 2\mathbf{i} + \mathbf{j} + 4\mathbf{k}$

(i) Find the exact length of AF

(ii) Find the exact length of BF

(iii) Find the vectors $\overrightarrow{AF}$ and $\overrightarrow{FB}$.

5 Areas of triangles & parallelograms

(a) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} -2 \\ 3 \\ 2 \end{pmatrix}$. Find the area of triangle OAB.

(b) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$. Find the area of triangle OAB.

(c) Relative to an origin O, the position vectors of the vertices A, B and C of a triangle are

$$\mathbf{i}+\mathbf{j}-\mathbf{k},\ 2\mathbf{i}+\mathbf{k} \text{ and } -\mathbf{i}+2\mathbf{j}-\mathbf{k}$$

respectively. By considering the displacement vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$, find

(i) the exact area of triangle ABC, and

(ii) the exact length of the vector component of $\overrightarrow{AB}$ perpendicular to $\overrightarrow{AC}$.

(d) The points A and B have coordinates $(5, 0, 5)$ and $(10, -5, 15)$ respectively. The point D is such that $OADB$ is a parallelogram and the point C divides AD in the ratio $2:3$. Find the exact areas of

(i) the parallelogram $OADB$, and

(ii) the quadrilateral $OCDB$.

6 For the following statements, state whether it is **true** or **false**.

(a) $\mathbf{a}\times\mathbf{a}=\mathbf{0}$.

(b) $\mathbf{i}\times\mathbf{j}=\mathbf{k}$.

(c) $\mathbf{a}\times\mathbf{b}=\mathbf{b}\times\mathbf{a}$.

(d) $\mathbf{a}\times\mathbf{b}=-(\mathbf{b}\times\mathbf{a})$.

(e) $\mathbf{a}\times\mathbf{b}$ is a scalar.

(f) $\mathbf{k}\times\mathbf{j}=\mathbf{i}$.

(g) $\begin{pmatrix}1\\2\\3\end{pmatrix}\times\begin{pmatrix}2\\4\\6\end{pmatrix}=\mathbf{0}$.

(h) $\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$.

(i) $(\lambda\mathbf{a})\times\mathbf{b}=\lambda(\mathbf{a}\times\mathbf{b})$ for all $\lambda\in\mathbb{R}$.

(j) $\mathbf{a}\times\mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

(k) $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{b}=0$.

(l) $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$ is a scalar.

(m) $|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}|\sin\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$.

(n) $|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}|$.

(o) If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$.

(p) For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ means that $\mathbf{a}$ is parallel to $\mathbf{b}$.

(q) The area of the parallelogram with adjacent sides $\mathbf{a}$ and $\mathbf{b}$ is $|\mathbf{a}\times\mathbf{b}|$.

(r) The area of triangle OAB is $|\overrightarrow{OA}\times\overrightarrow{OB}|$.

(s) $\mathbf{a}\times(\mathbf{b}\cdot\mathbf{c})$ is a vector.

(t) For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}|$, then $\mathbf{a}$ is perpendicular to $\mathbf{b}$.

(u) $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$.

(v) $|\mathbf{a}\times\mathbf{b}|^2+(\mathbf{a}\cdot\mathbf{b})^2=|\mathbf{a}|^2|\mathbf{b}|^2$.

(w) $\mathbf{a}\times\mathbf{b}=\mathbf{a}\times\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$.

(x) $\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$.

(y) The shortest distance from B to the line through A parallel to $\mathbf{d}$ is $|\overrightarrow{AB}\times\mathbf{d}|$.

(z) If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ and $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$.

END OF WORKSHEET